

Milestone 1 Showcase & Presentation Guide

Project: ResearchSphere AI - AI-Powered Research Funding & Innovation Intelligence Platform

1. DEFAULT PRE-SEEDDED CREDENTIALS & LOGIN

- **Administrator Email:** admin@researchsphere.ai
- **Administrator Password:** Admin@123456
- **Pre-Seeded Role:** Administrator (Access to Admin Console, Users Table, Audit Logs)
- **App Web URL:** http://localhost:5173
- **Backend Swagger API Docs:** http://localhost:8000/docs

2. HOW TO REGISTER A NEW ACCOUNT (SIGN UP)

To demonstrate user registration and Role-Based Access Control (RBAC):

- **Step 1:** Open http://localhost:5173/register or click "Get Started" on the Landing Page.
- **Step 2:** Enter Full Name (e.g. Dr. Alex Rivera), Email (alex@university.edu), and Password.
- **Step 3:** Click one of the 4 Role Selector Buttons: Researcher, Startup Founder, Innovation Manager, Administrator.
- **Step 4:** Click "Register Account" -> Auto-creates account & profile, issues JWT token, redirects to Role Dashboard.

3. STEP-BY-STEP LIVE SHOWCASE SCRIPT (5-7 MINUTES)

Step 1: Landing Page Overview (1 Min)

Show http://localhost:5173. Highlight the dark glassmorphism styling, live metric counters (250M+ Publications, 140M+ Patents, \$15B+ Grants), and 4 Platform Innovation Pillars.

Step 2: Admin Console & Dual-Database Audit Stream (1 Min)

Sign in as admin@researchsphere.ai / Admin@123456. Open Admin Console (/admin). Show User Accounts table, platform metrics, and switch to System Audit Logs tab showing live event tracking (USER_LOGIN, REGISTER, PROFILE_UPDATE).

Step 3: Research Profile Management (1 Min)

Register a new Researcher account. Open Profile (/profile). Add professional title, Research Domains (Artificial Intelligence, Quantum Computing), and Technology Keywords (#deeplearning, #drugdiscovery). Click Save Changes and show live rendered tag pills.

Step 4: Live Publication Dataset Integration (1.5 Mins)

Open Publications (/publications). Search query "artificial intelligence". Filter by source (OpenAlex, CrossRef, Semantic Scholar). Showcase live paper cards with title, authors, venue, year, citation count, and DOI external links.

Step 5: Live Patent Intelligence Explorer (1 Min)

Open Patents (/patents). Search "quantum computing". Filter by source (USPTO, Google Patents, The Lens). Showcase patent numbers, assignees, and status badges (Granted / Pending).

4. TECHNICAL ARCHITECTURE & TALKING POINTS

- **Hybrid Database Architecture:** PostgreSQL 16 handles 11 relational tables (Users, Profiles, Pubs, Patents, Audit Logs) for strict ACID compliance; MongoDB 7 handles raw API payload caching.
- **Backend Clean Architecture:** Built with FastAPI following 4-tier separation: Routers -> Services -> Repositories -> Models.

ResearchSphere AI - Milestone 1 Showcase & Presentation Guide

- **Security & RBAC:** Bcrypt salted password hashing, JWT OAuth2 bearer security, and role enforcement guards (require_role).

- **Zero-Dependency Resilience:** Automatic SQLite fallback allows the app to run locally with zero external database dependencies.